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Pet grooming blade storage tray

Abstract

A storage tray for clipper and trimmer cutting blades for use in pet grooming. The storage tray is securable to a vertical arm of a conventional pet grooming table allowing the table top to remain obstacle free. The storage tray includes a fan for faster heat dissipation from hot cutting blades. An extension tray can be coupled to the storage tray with receptacles for scissors, clippers and the like grooming tools. A further receptacle is constructed and arranged to hold a cell phone.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) In accordance with 37 C.F.R. 1.76, a claim of priority is included in an Application Data Sheet filed concurrently herewith. Accordingly, the present invention claims priority as a CIP application to U.S. patent application Ser. No. 18/155,125 entitled “PET GROOMING BLADE COOLING TRAY” with a filing date of Jan. 17, 2023 which claims priority to U.S. Provisional Patent Application No. 63/300,087 filed Jan. 17, 2022, entitled “PET GROOMING BLADE COOLING TRAY,” the contents of the Applications are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The invention is directed to the field of pet grooming and, in particular, to a storage tray for pet grooming clipper and trimmer blades.

BACKGROUND OF THE INVENTION

(2) Pet grooming appliances, commonly referred to as clippers and/or trimmers, include interchangeable cutting blades for clipping and trimming pet hair. The interchangeable cutting blades permit the use of different types of cutting blades to treat the hair types found with various breeds of animals.

(3) Skip tooth blades are a standard option to start a grooming procedure while fine-teeth blades have closely spaced teeth ideal for thin and soft haired pets. Blades can be further sized for coat length. The type of blade and blade size allow the pet groomer to carefully groom the pet to beyond the pet owner's expectations.

(4) Blade construction may be chrome plated stainless steel, carbon, silver or titanium. Cutter blade sizes can be expressed in numbers from 3 to 50. For instance: number 3 blades may provide a 13 mm coat length; number blades 5 may provide a 6 mm coat length; number 7 blade may provide a 3.2 mm coat length; number 9 blade may provide a 2 mm coat length; number 30 blade may provide a 0.5 mm coat length to name a few. Commonly it takes 2-4 blades to groom one dog. For instance, Poodle groomers would likely use a number 7 blade for the body, number 10 or 15 for the

head, number 3 or 5 for the legs, and number 10 or 15 for the tail. During the grooming process, the cutting blade will heat up and, if left unchecked, can burn the skin of the animal being groomed. For this reason blade changes are frequently made allowing a cutting blade to cool off during use. Preferably the various interchangeable blades are kept near the grooming table so that the pet being groomed is not left unattended.

(5) Thus, a need exists for a storage tray securable to a grooming table for use in dissipating the heat from cutting blades using a platform elevated above the grooming table.

SUMMARY OF THE INVENTION

(6) Disclosed is a storage tray having a plurality of slanted surfaces, each configured to support a trimmer blade. Upon resting a hot blade against the surface, a cooling fan passes air through the storage tray to dissipate heat quickly and permit the interchangeable use of blades during grooming sessions. Securing the storage tray to a grooming table pole vertical pole keeps the grooming table free of obstructions. The cooling fan is powered through a USB cable coupling to an external power source.

(7) In the instant invention, the storage tray has a generally rectangular main body constructed of plastic, the bottom of the storage tray having a plurality of through holes. The tray has a plurality of separate sections, each section having a slanted surface upon constructed and arranged to hold an individual cutting blade. In another embodiment, marble, ceramic, copper or the like heat dissipating material can be placed over the slanted surface to provide enhanced heat dissipation. The storage tray is attached to the vertical arm of a grooming table by clamping mechanism to adjust the height of the storage tray in relation to the grooming table. When in use, the storage tray is easily accessible by the groomer for insertion and retrieval of specific cutting blades during a pet grooming session. An extension tray can be attached to the storage tray to hold clippers, scissors and the like items commonly used during the grooming process. A cell phone holder is formed integral to the clamping mechanism.

(8) An objective of the invention is to provide a support tray for clipper and trimmer blades that is securable to the vertical arm of a conventional grooming table.

(9) Another objective of the invention is to provide a support tray containing cooling fans for use in cooling clipper and trimmer blades.

(10) Yet still another objective of the invention is to provide an economical support tray that can be sized to fit four or more cutting blades allowing ease of access to the pet groomer and assuring cutting blades are cooled quickly allowing reuse.

(11) Still another objective is to provide an extension tray for holding of pet grooming devices. The storage tray and extension tray are releasably secured to a clamping mechanism, the clamping mechanism is attached to the vertical pole found on professional grooming tables.

(12) Other objectives and advantages of this invention will become apparent from the following description taken in conjunction with any accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention. Any drawings contained herein constitute a part of this specification and include exemplary embodiments of the present invention and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a perspective view of the blade tray as installed on the vertical arm of the grooming table;

(2) FIG. 2 is a first side view of the blade tray as installed on the vertical arm of the grooming table;

(3) FIG. 3 is a top view of the blade tray;

- (4) FIG. 4 is a top perspective view of the blade tray as installed on the vertical arm of the grooming table;
- (5) FIG. 5 is perspective view of the blade tray installed on a grooming table;
- (6) FIG. 6 is a perspective view of a blade and extension tray;
- (7) FIG. 7 is an exploded view thereof;
- (8) FIG. 8 is a upper perspective view of the blade tray;
- (9) FIG. 9 is a lower perspective view of the blade tray;
- (10) FIG. 10 is a top right perspective view of a blade and extension tray with a telephone holder;
- (11) FIG. 11 is a top left perspective view thereof;
- (12) FIG. 12 is a top view thereof;
- (13) FIG. 13 is a bottom perspective view thereof;
- (14) FIG. 14 is a top perspective view of a blade holder;
- (15) FIG. 15 is a bottom perspective view of the blade holder.

DETAILED DESCRIPTION OF THE INVENTION

(16) While the present invention is susceptible of embodiment in various forms, there is shown in the drawings and will hereinafter be described a presently preferred, albeit not limiting, embodiment with the understanding that the present disclosure is to be considered an exemplification of the present invention and is not intended to limit the invention to the specific embodiments illustrated.

(17) Now referring to FIG. 1-4, illustrated is one embodiment of the present invention and the manner in which it is used, with like components numbered consistently throughout. Disclosed is a grooming blade storage tray **10** formed from a rectangular frame member preferably rigid plastic. Aluminum or other lightweight rigid material may also be used. The storage tray **10** having a front wall **12**, a rear wall **14**, a first side wall **16**, and a second side wall **18**. The second side wall **18** second forming an access door **15**. Between the first side wall **16** and second sidewall **18** are slanted assembly support members **20** constructed and arranged to support a thin sheet of marble material **22** on a leading surface **24** of the support member **20**, the trailing surface **26** may be used to place trimmers have cooled so that a blade that is cooling down on the marble material **22** is not accidentally placed back into the surface. Preferably the walls and slanted blade storage sections **12** are comprised of plastic. It is noted that the illustrated tray **10** in the figures has four sections. To ease in drawing clarity only a single section **12** is enumerated, the remaining sections forming a mirror image thereof. A tray with 2-8 sections **12** is contemplated to be within the scope of this invention. The tray **10** is mounted to the vertical arm **50** of the grooming table, see FIG. 10. The placement of the marble material **22** on the slanted surface in each section **12** may be opposite on each side of the vertical arm for ease of accessing the cutter blades.

(18) FIG. 2 illustrates sidewall **18** with access door **15** secured to the sidewall **18** by fasteners **30**, **32**. The access door **15** is temporarily removed from the sidewall **18** allowing access to cavity **34** within the tray **10**. The cavity can accommodate up to a one inch square vertical arm **50** by use of the appropriate foam rubber spacer **36** providing a friction fit wherein the access door is fastened to the sidewall to engage all sides of the vertical arm **50**.

(19) FIG. 3 is an upper view of the storage tray **10** depicting the cavity for receipt of the vertical arm, not shown. The foam rubber spacer **36** depicted can be easily compressed with the attachment of the access door **15**, sufficient to support the weight of the tray **10** and any blades place thereon. In the preferred embodiment the rubber space **36** is placed on two inner walls **40** and **42**, to assure alignment with the vertical arm **50** along inner side walls **44** and **46**. FIG. 4 depicts the storage tray **10** with the access door **15** removed from the sidewall **18** allowing attachment to, or removal from, the vertical arm **50**.

(20) FIG. 5 depicts a conventional grooming table **52** with a vertical arm **50** extending upwardly from the table **52** to a position above the animal's frame **60**. The vertical arm is commonly used to harness the pet in a stance on the table. In this illustration a second vertical arm **54** is used in

combination with the first vertical arm **50** to hold up a horizontal bar **56** used to position the animal's frame **60** in a properly controlled position for grooming using alignment straps **57** and **58**. (21) Referring disclosed is an now to FIGS. **6-9** disclosed is an embodiment that allows cooling by use of a marble or the like heat sink **85**, and/or by use of a cooling fan **100**. In this embodiment the storage tray **70** is a frame member **72** having a support plate **74** with an upper surface **76** and a lower surface **78** forming a thickness therebetween, the thickness depending upon the material employed with sufficient thickness to provide for the rigidity. The frame member **72** has a length generally about 270 mm and a width of generally about 124 mm. The support plate **74** has a plurality of vent holes **80** extending between the upper **76** and lower **78** surface. The frame member **72** further having a front side wall **82** spaced apart from a rear side wall **84**, and a left side wall **86** spaced apart from a right side wall **88**. The side walls extending upwardly from the upper surface **76** of the support plate **74**. In the preferred embodiment the frame member **72** is constructed from plastic or aluminum. The frame member **72** having a shape that allows manufacturing by use of a 3D printer when constructed of plastic.

(22) The support plate **74** includes a plurality of slanted tray surfaces **90** positioned between the left side wall **86** and the right side wall **88**. The slanted surface **90** constructed and arranged to receive a cutting blade, not shown. In one embodiment the slanted tray surface **90** may include a marble, ceramic, copper or the like heat dissipating heat sink **85** placed over a slanted tray surface **90** which operates to dissipate the cutting blade heat. In operation the groomer may have multiple cutting blades placed on the slanted tray surfaces **90** wherein the cutting blades can be changed out so that the pet is not subjected to a hot blade. Conventional cutting blades use a mechanical shearing action resulting in friction that produces appreciable heat and can burn the pet. Frequent changing of the cutting blades is seen as a benefit to the welfare of the pet. A cooling fan **100** is secured to the lower surface **78** of the floor section. The cooling fan **100** electrically coupled to a power source and used to force air through the vent holes **80** to assist in dissipating heat from cutting blades placed on each slanted surface **90**. The cooling fan **100** includes a USB plug **102** for electrically coupling to a power source having a USB port. The power source can be an electrical outlet or a portable rechargeable battery.

(23) A slide member **110** for securing the frame member **72** to the vertical grooming arm **180** of a pet grooming table. Attachment can be with fasteners or tabs for interlocking. The slide member **110** having at least one aperture **112** constructed and arranged to be slidably positioned over the vertically disposed grooming arm **150**. A bracket member **120** comprising a first element **122** releaseably coupled to a second element **124** using fasteners **126** is positioned around the grooming arm wherein the bracket member **120** frictionally engages the grooming arm and operates as a platform to maintain the storage tray **70** at a height above the grooming table.

(24) In a previous embodiment the clamp mechanism is defined as a slide member attached to the rear wall of the frame member as illustrated in the previously described embodiment FIGS. **1-5**. In this embodiment the clamp mechanism is defined as cavity bounded between the rear wall and a slanted surface, and the left side wall and the right side wall. A portion of the right side wall, access door, is releaseably attached forming an access door, the cavity including at least one compressible foam rubber spacer to frictionally engage the vertical grooming arm for securing the frame member thereto. The foam rubber spacer compressed against the vertical arm and secured in position by attachment of the access door. In another embodiment, a leg, not shown, may extend downward from a lower surface of the support plate, floor for securement by use of a hand screw, wherein the hand screw may be used to frictionally engage the vertical arm for securement of the frame member to the vertical arm.

(25) An extension tray **130** is securable to the slide member **110** to provide a further storage area to support the grooming procedure. Similar to the storage tray **70**, the extension tray **130** may be secured to the slide member **110** by fasteners or tabs for interlocking. The extension tray **130** has a length of generally about 290 mm and a width of generally about 124 mm. The extension tray **130**

has a plurality of openings **132** sized for receipt of scissors, not shown and openings **134** for trimmers, not shown. Two openings **132** have a dimension of generally about 38 mm by 50.8 mm for use in holding scissors. Three openings **134** have a dimension of generally about 25.4 mm square for use in holding clippers. A circular opening **136** has a diameter generally about 100 mm and is used as a holder for a cup, mug or miscellaneous tray, not shown. A cell phone receptacle **140** is provided to hold a conventional sized cell phone, the receptacle **140** having a width of generally about 25.4 mm and a length of generally about 120 mm.

(26) In one embodiment the cooling fan **100** includes a USB receptacle **152** that is electrically coupled to the fan's USB inlet plug **102** wherein a cell phone, not shown, placed in the receptacle **140** can be attached to the cooling fan USB receptacle **152** for recharging of a cell phone from the cooling fan's USB inlet port **102**. The storage tray **70** may be directly placed on a table with non slip bases **154** to maintain a distance between the fans and a table. Similarly the storage tray **70** may be secured to a wall using fasteners placed through wall hanging slots **156**. In the preferred embodiment, two cooling fans **100** are employed having a substantially square shape wherein placement in a side by side position allows for a rectangular shape of the tray **70**.

(27) Referring now to FIGS. **10-15** disclosed is an embodiment similar to that illustrated in FIGS. **6-9** having a cooling fan **200**. In this embodiment the storage tray **270** is a frame member **272** having a support plate **274** with an upper surface **276** and a lower surface **278** forming a plate having a thickness, the thickness depending upon the material employed with sufficient thickness to provide for the rigidity. In one embodiment the frame member **272** has a length generally about 270 mm and a width of generally about 124 mm. The support plate **274** has a plurality of vent holes **280** extending between the upper **276** and lower **278** surface. The frame member **272** further having a front side wall **282** spaced apart from a rear side wall **284**, and a left side wall **286** spaced apart from a right side wall **288**. The side walls extending upwardly from the upper surface **276** of the support plate **274**. In the preferred embodiment the frame member **272** is constructed from plastic or aluminum. The frame member **272** having a shape that allows ease of manufacturing, even by use of a 3D printer.

(28) The support plate **274** includes a plurality of slanted tray surfaces **290** positioned between the left side wall **286** and the right side wall **288**. The slanted surface **290** constructed and arranged to receive the cutting blades, not shown. In operation the groomer may have multiple cutting blades placed on the slanted tray surfaces **290** wherein the cutting blades can be changed out so that the pet is not subjected to a hot blade. Frequent changing of the cutting blades is seen as a benefit to the welfare of the pet. The cooling fan **100** is secured to the lower surface **278** of the support plate **274**. The cooling fan **100** electrically coupled to a power source and used to force air through the vent holes **280** to assist in dissipating heat from cutting blades placed on each slanted surface **290**. As with the previous embodiment, the cooling fan **100** includes a USB plug **102** for electrically coupling to a power source having a USB port. The power source can be an electrical outlet or a portable rechargeable battery.

(29) A slide member **210** for securing the frame member **272** to the vertical grooming arm **150** of a pet grooming table. The slide member **210** having at least one aperture **212** constructed and arranged to be slidably positioned over the vertically disposed grooming arm **150**. A bracket member **220** can include friction inducing tabs **222**, **224** to engage the grooming arm **150** to stop the slide member **210** from descending the grooming arm **150** which operates as a platform to maintain the storage tray **270** at a height above the grooming table. A receptacle **231** is provided to hold a conventional sized cell phone. Placement of the receptacle **231** on the side edge allows ease of viewing from most every position on the grooming table.

(30) In a previous embodiment the clamp mechanism is defined as a slide member attached to the rear wall of the frame member as illustrated in the previously described embodiment FIGS. **1-5**. In this embodiment the clamp mechanism is defined as cavity bounded between the rear wall and a slanted surface, and the left side wall and the right side wall. In another embodiment, the friction

inducing tabs can be threaded thumb screws that can be tightened to maintain the storage tray height.

(31) An extension tray **230** is securable to the slide member **210** to provide a further storage area to support the grooming procedure. In one embodiment, the storage tray **270** and extension tray **230** hook to the slide member **210** allowing attachment and removal without tools. The extension tray **230** has a length of generally about 290 mm and a width of generally about 124 mm. The extension tray **230** has a plurality of openings **232** sized for receipt of scissors, not shown and openings **234** for trimmers, not shown. Two openings **232** have a dimension of generally about 38 mm by 50.8 mm for use in holding scissors. Three openings **234** have a dimension of generally about 25.4 mm square for use in holding clippers. A circular opening **236** has a diameter generally about 100 mm and is used as a cup holder, not shown. A receptacle **240** is provided for a cell phone or any other item, the receptacle **240** having a width of generally about 25.4 mm and a length of generally about 120 mm.

(32) In accordance with the previous embodiment the cooling fan **200** includes a USB receptacle that is electrically coupled to the fan's USB inlet plug wherein a cell phone, not shown, placed in the receptacle **231** can be attached to the cooling fan USB receptacle for recharging of a cell phone from the cooling fan's USB inlet port. The storage tray **270** may be directly placed on a table with non slip bases **254** to maintain a distance between the fans and a table. Similarly the storage tray **270** may be secured to a wall using a hanging slot.

(33) It is to be understood that while a certain form of the invention is illustrated, it is not to be limited to the specific form or arrangement herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention, and the invention is not to be considered limited to what is shown and described in the specification and any drawings/figures included herein.

Claims

1. An air cooled storage tray for a pet grooming table comprising: a clamp mechanism releasably securable to a vertically disposed grooming arm of a pet grooming table; a storage tray releasably securable to said clamp mechanism, said storage tray having a rectangular frame member formed from a support plate having an upper surface and a lower surface with a plurality of vent holes extending therebetween, said frame member having a front side wall spaced apart from a rear side wall, and a left side wall spaced apart from a right side wall, each of said side walls extending upwardly from said upper surface of said support plate; a plurality of slanted tray surfaces positioned between said left and right sidewall and above said upper surface of said frame member, each said slanted tray surface constructed and arranged to receive a cutting blade; an extension tray securable to said clamp mechanism, said extension tray including a plurality of openings for receipt of at least one scissor and at least one trimmer; and at least one cooling fan secured to said lower surface of a floor section, said cooling fan electrically coupled to a power source and used to force air through said vent holes to assist in dissipating heat from each said cutting blade placed on each said slanted tray surface; wherein said clamp mechanism is used for positioning said storage tray at an elevation on said grooming arm and at least one cooling fan used to cool cutting blades placed on said slanted tray surfaces.
2. The storage tray according to claim 1 wherein said clamp mechanism includes a receptacle constructed and arranged to receive a cellular telephone.
3. The storage tray according to claim 1 wherein at least one of said openings has a dimension of generally about 38 mm by 50.8 mm for use in holding a nail clipper.
4. The storage tray according to claim 1 wherein three of said openings having a dimension of generally about 25.4 mm square for use in holding scissors.
5. The storage tray according to claim 1 including a circular opening having a diameter generally

about 100 mm, wherein said circular opening can hold a coffee mug.

6. The storage tray according to claim 1 wherein said cooling fan includes a Universal Serial Bus plug for electrically coupling to a power supply.

7. The storage tray according to claim 1 wherein said cooling fan includes a Universal Serial Bus receptacle electrically coupled to a Universal Serial Bus plug wherein said Universal Serial Bus receptacle is available to receive a charging cord from a cell phone.

8. The storage tray according to claim 1 including a receptacle integral to said extension tray for a cell phone, said receptacle having a width of generally about 25.4 mm and a length of generally about 120 mm.

9. The storage tray according to claim 1 wherein each slanted tray surface includes a heat sink selected from the group of: marble, ceramic, or copper, for use in dissipating heat upon contact with said cutting blade.

10. The storage tray according to claim 1 wherein said frame member is constructed from plastic or aluminum.
